Solana

DIY Instructions

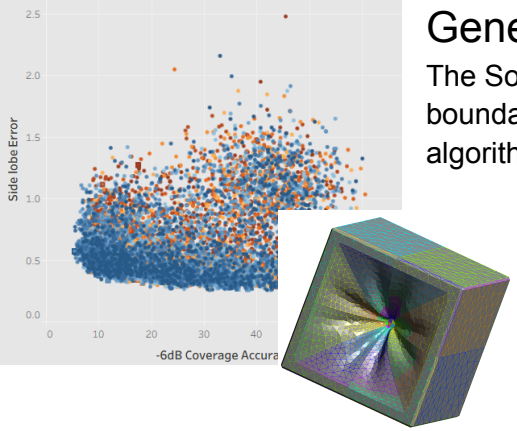


The Solana is a versatile point source loudspeaker featuring flexible uses in DJ booth monitoring, fill, domestic, and home theater applications.

JW SOUND

Design Overview

This is a compact 2-way top that prioritizes smooth, well controlled directivity, portability, and moderate output in a symmetric package.



Genetic Algorithm Shape Optimization

The Solana platform represents a case study in algorithmic loudspeaker shape optimization using the boundary element method. The waveguide profile was iterated and developed via an evolutionary algorithm to minimize off-axis lobing and polar irregularities, while achieving pattern control down to the low-midrange. Version 1.0 was the result of a focused optimization campaign with a basic genetic algorithm script that produced 350 candidates over 7 generations. Version 2.0 is the product of a more advanced GA that simulated over 4,400 unique shapes before arriving at the final geometry.

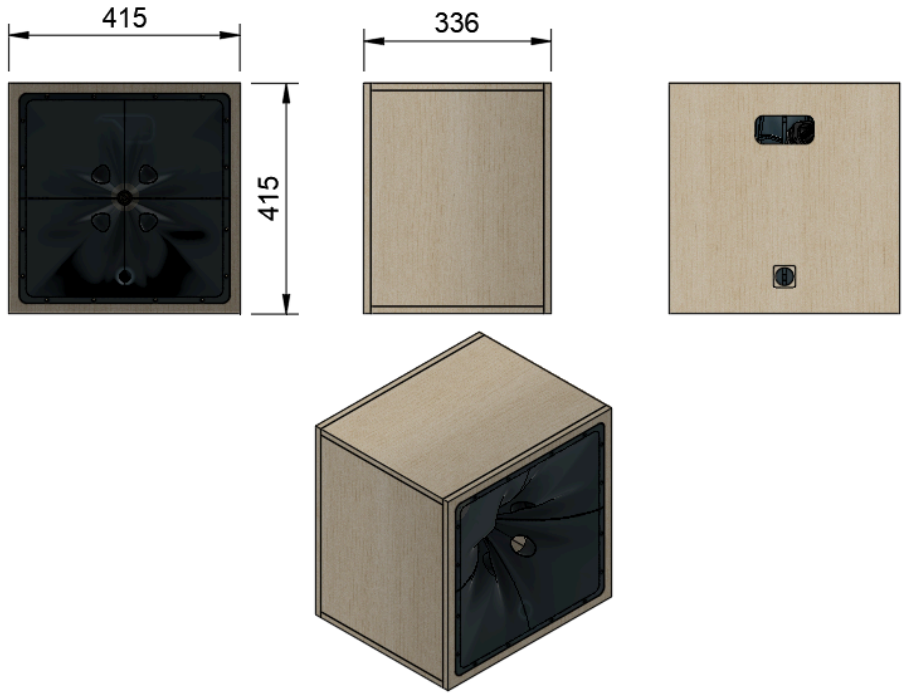
Enclosure Design

The provided enclosure plans specify a simple ~25 liter rear chamber tuned at 70Hz with a single rear port that serves as a carrying handle. The quad 6.5” woofers are effectively loaded into a 6th order bandpass system that is best suited for crossing over to subs around 100Hz using a 4th order acoustic slope. However the enclosure can be modified by lengthening the port for full-range use at lower SPLs. A target rear chamber tuning frequency of 50Hz is recommended for low-tuned enclosure modifications.

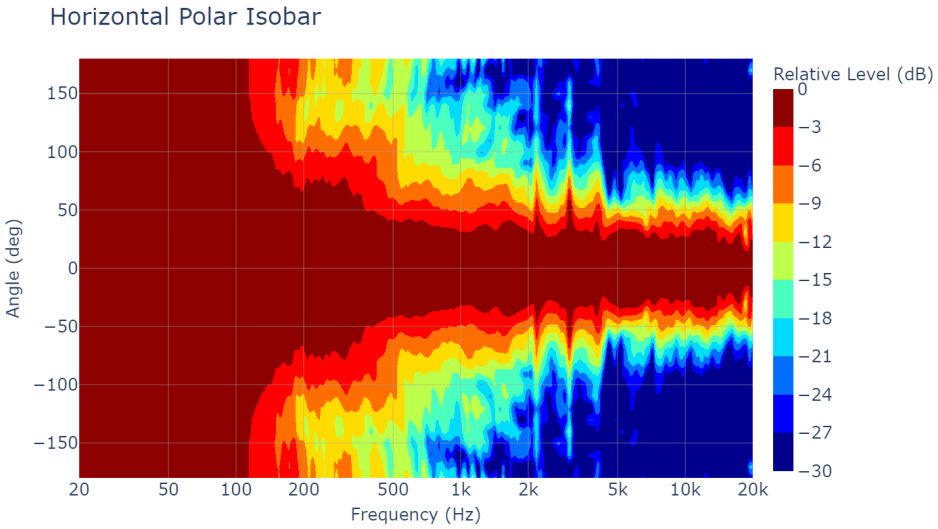


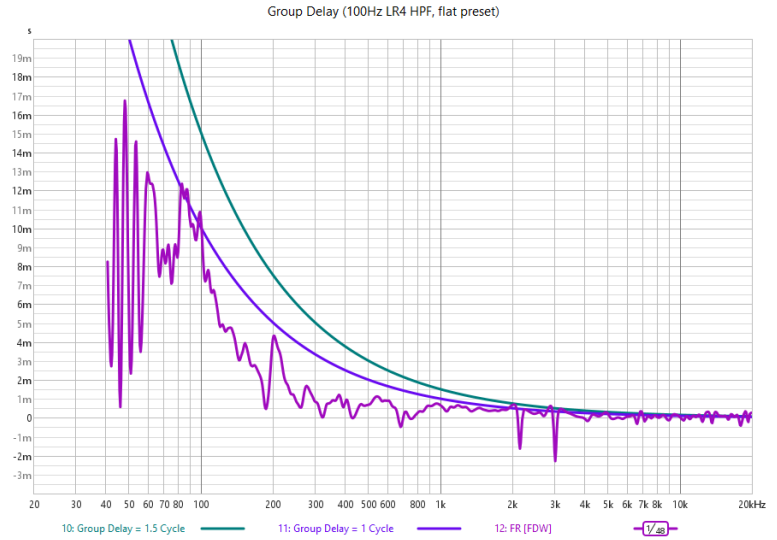
Specifications

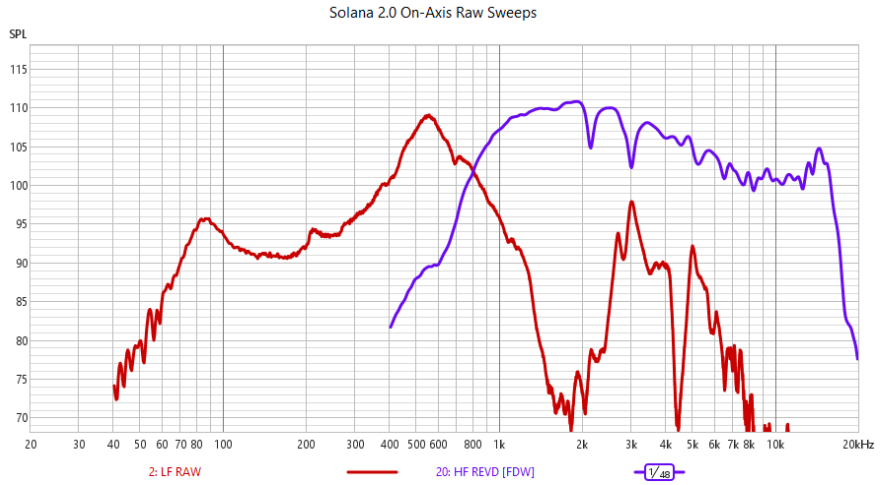
Dimensions: 41cm x 41cm x 34cm
(16” x 16” x 13”)
Weight: 12.5KG (27.5LBS)
Horizontal Directivity: 90° constant directivity
Vertical Directivity: 90° constant directivity (symmetric)
Acoustic Crossover Frequency: 1200Hz LR6
Maximum Output: See THD limited max SPL on page 4
Drivers: 4x 6.5” midrange drivers, 1x 1” compression driver
Construction Material: 18mm birch plywood
Amplification: Bi-amplified with DSP
Recommended Amplification: 500W+ for LF, 80W+ for HF

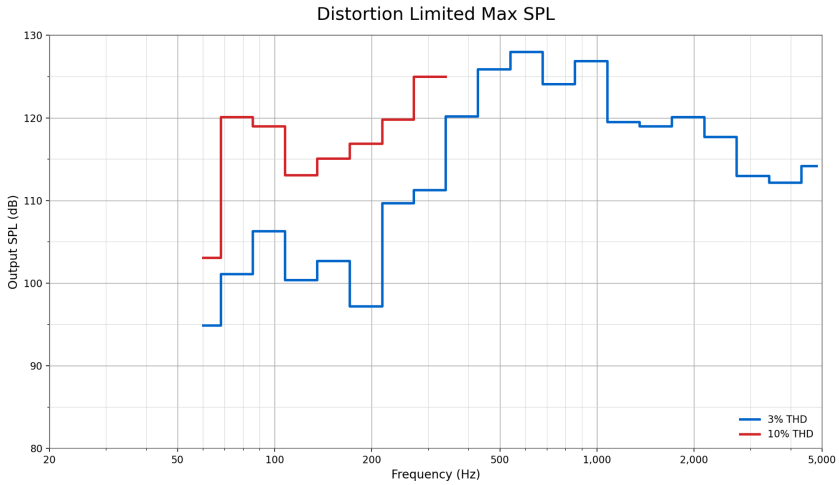


Measurements Ground plane, 1/48 smoothing, 30 cycle FDW









Distortion limited max SPL is calculated via ⅓ octave stepped sine tone bursts with 683ms length for frequencies <300Hz and 171ms length for frequencies >300Hz. A 2,000ms rest period is used between bursts. The polar plot is normalized to the design's 10 degree reference axis

Assembly Introduction & Requirements

Download Required Files HERE

Drivers (for each waveguide)

Type	Model	Notes
LF	B&C 6NDL38	4Ω/8Ω/16Ω are all acceptable
HF	B&C DH450	16Ω version is preferred for lower noise floor, but 8Ω works fine

LF wiring note: 4x8-ohm LF drivers can be wired in series-parallel for a nominal 8-ohm load. 4x16-ohm drivers can be wired in parallel for a nominal 4 ohm load. 4x4 ohm drivers can be wired in series-parallel for a nominal 4 ohm load. Use the appropriate wiring strategy according to driver availability and amplifier capabilities.

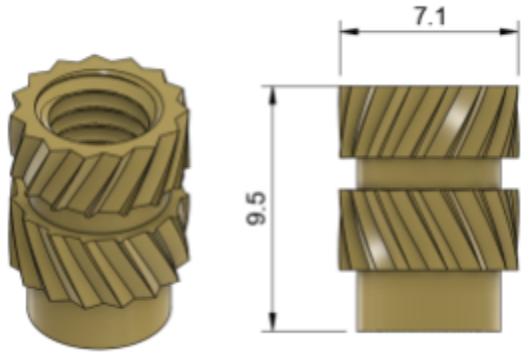
FDM Printing Requirements

Minimum print volume: 250x210x220mm for 4-piece assembly **OR** 340x320x340mm for 2-piece print.
Printer Enclosure: An enclosure is recommended to minimize warping during printing.
Material: Consider the environment you will keep your speaker in long-term as it relates to temperature changes and mechanical stress. Carbon or glass fiber-filled filaments are recommended as they help the parts maintain dimensional stability during printing. Epoxy or fiberglass coatings can be applied on the back side of the waveguide module after assembly for additional strength across layer lines. TPU is used for the optional midrange mounting gasket.

Best: PET-CF/GF, PPA-CF/GF, PPS-CF/GF, PAHT-CF, PA6/12-CF/GF, PEEK
Good: PETG-CF/GF, PA6/12*, ABS-CF/GF, ASA-CF/GF, PC
Acceptable: ASA*, ABS*, PETG*, PCTG
Avoid: PLA, TPU (for structural pieces)

Hardware, Tools, and Fasteners (for 1 unit)

Please follow the hardware type and quantity carefully for best results.



A. **16x** M5 x 20mm button cap fastener
B. **16x** M5 heat set threaded insert (important - must be 7.1mm outside diameter)
C. **2x** M5 x 12mm countersunk fastener
D. **20x** M4 threaded insert (must be 6.0mm outside diameter)
E. **16x** M4 x 12mm button cap fastener
F. **16x** M4 x 20mm button cap fastener
G. 16x M4 tee nut or furniture insert
H. 3x M6 tee nut or furniture insert
I. 3x M6 x 20mm countersunk fastener
J. **1x** Speakon NLT4MPXX or NLT4MPXX-BAG receptacle
K. **1x** Penn Elcom M1557/M20 Universal Speaker Mounting Top Hat

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- ½ sheet of 1220x2440x18mm (4’x8’x3/4”) plywood - baltic birch is preferred
- Adhesive for assembling 3D printed pieces - two part epoxy for plastic is recommended
- Soldering iron for threaded inserts
- Sandpaper
- Wood glue or construction adhesive for cabinet assembly
- Assorted wood clamps (8-12 are recommended)
- Hex keys, spanners, and screwdrivers as needed
- Measurement microphone and audio interface for DSP tuning
- 1.5 - 2.0mm (14-16 gauge) internal speaker wiring
- 10mm (⅜”) wide gasket tape
- 2.5cm (1”) thick polyester enclosure lining
- Brad nails or cabinet screws

DSP Settings

Important note - for designs such as the Solana, it's critical to state that the electrical crossover points and slopes (the ones configured in DSP) are not always equal to the acoustic crossover points and slopes (what the drivers are actually producing). For best results, configure target acoustic slopes within your measurement software, and then use a combination of gain, parametric EQs, and electrical LPFs/HPFs to align the measured response until it matches the targets.

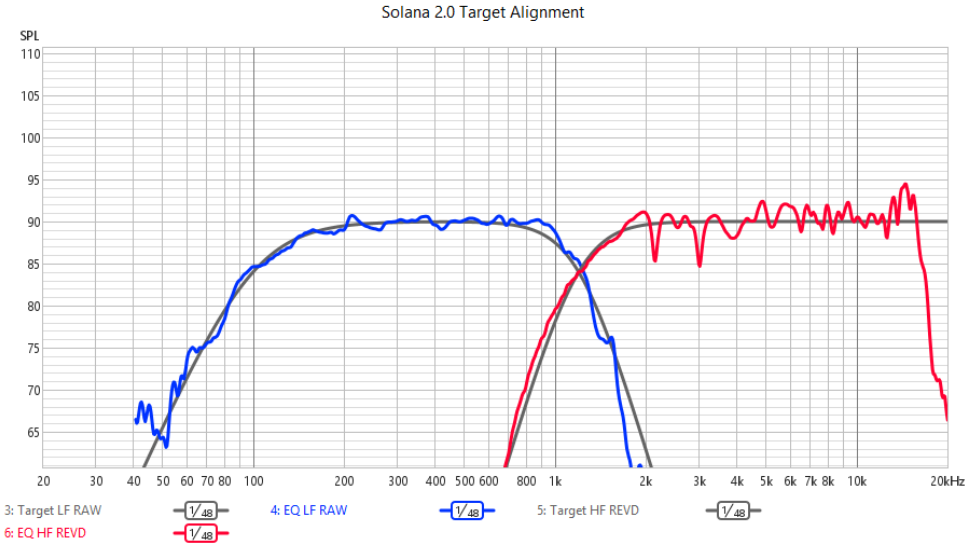
The following DSP settings are provided as target curves for tuning. The Loudspeaker Measurement & System Alignment Guide is a comprehensive resource that can be referenced for measuring and aligning the Solana.

Channel 1 (Woofers)

Target Acoustic LPF: 1200Hz LR 36dB/octave
Target Acoustic HPF: 100Hz LR 24dB/octave
Minimum Safe Electrical HPF: 70Hz LR 12dB/octave

Channel 2 (Compression Driver)

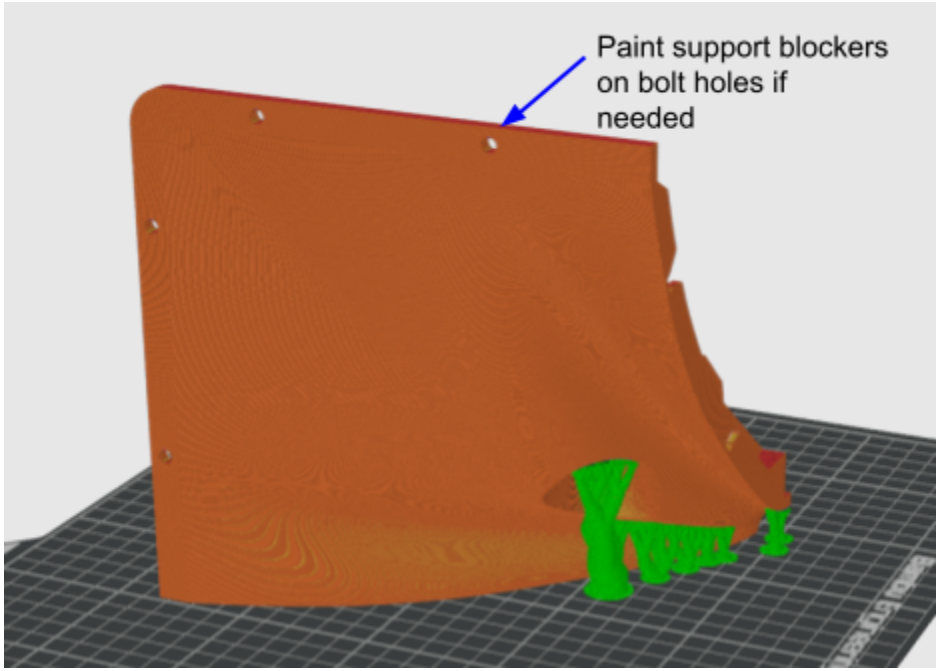
Target Acoustic LPF: None
Target Acoustic HPF: 1200Hz LR 36dB/octave
Minimum Safe Electrical HPF: 1000Hz LR 12dB/octave



Measure with a 10 degree microphone reference axis.

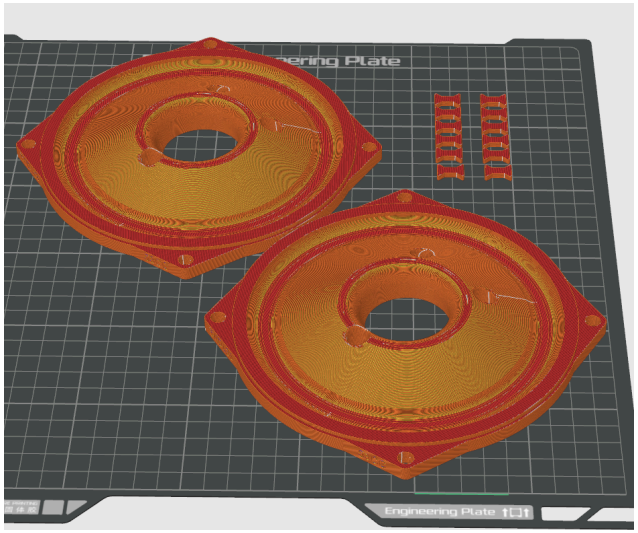
FDM Printing

Adjust settings as needed for your printing setup. The below settings are recommended as a starting point.



- 4x *quarter.stl (or 2x half.stl)*
- *4x Midflange.stl (35% infill)*
- 1x *throatadapter.stl (100% infill)*
- 12x *butterflypin.stl (100% infill)*

Waveguide Print Settings:
Infill: 35%
Infill Pattern: Adaptive cubic or cubic
Wall Loops: 3
Layer Height: 0.20mm
Supports: On build plate only. 40% overhang threshold. Use organic supports if available



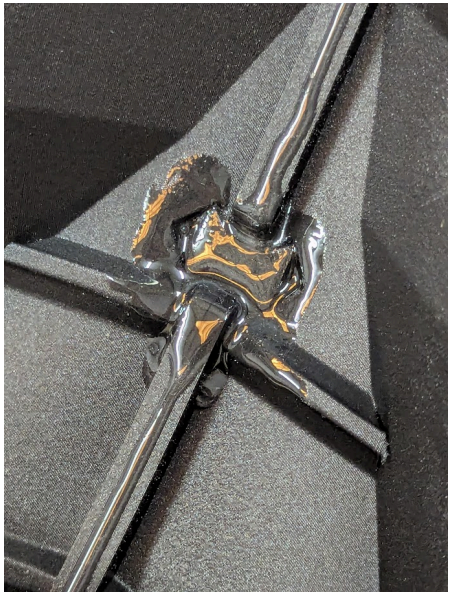
If printing the optional TPU midrange gaskets, use 100% infill.

Assembly

1. Sand the mating flanges of each waveguide piece with a medium grit sandpaper to allow for better adhesive bonding. Apply adhesive to the mating surfaces and join the waveguide pieces together. Insert the butterfly pins into their pockets and apply a liberal amount of adhesive around them. Place the joined pieces on a flat surface and use clamps to align the seams and allow time for the adhesive to set. Repeat this for all pieces of the waveguide.

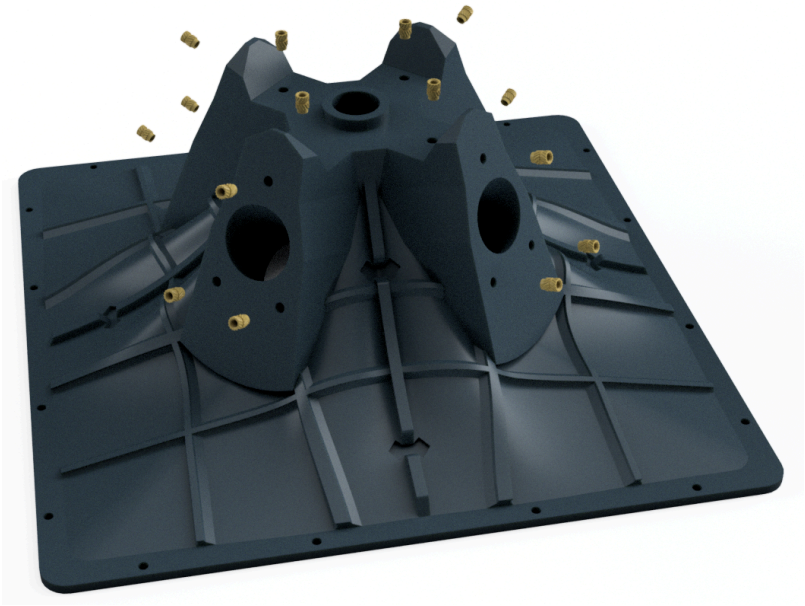


Example of butterfly pin epoxy potting and clamping strategy. A flat piece of scrap wood is used as a platform to apply downward clamping force for seam alignment





2. Use a soldering iron to insert the 16x M4 heat set inserts into the waveguide body.



3. Use a soldering iron to insert the 16x M5 heat set inserts into the midrange mounting flanges



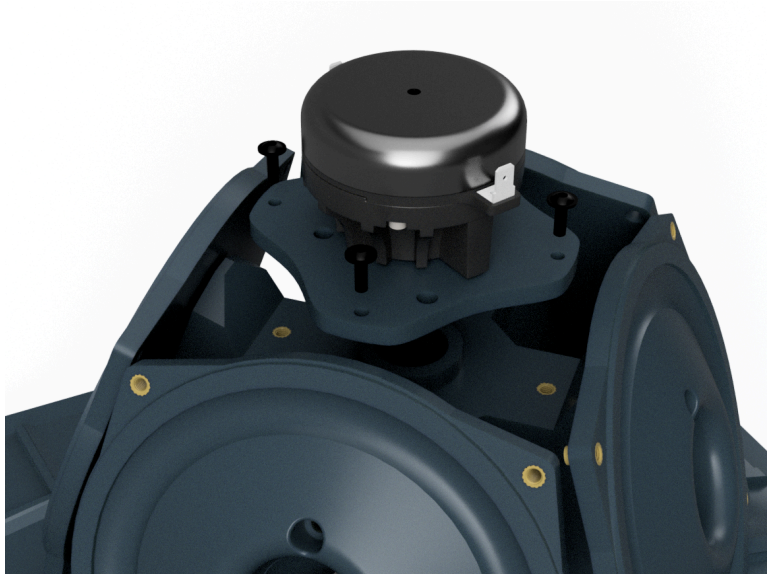
4. Attach the four midrange flanges to the waveguide body using M4x12mm fasteners. The flanges need to seal airtight against the waveguide body. You can either print the optional TPU gasket supplied in the files, or apply a ring of foam gasket tape.



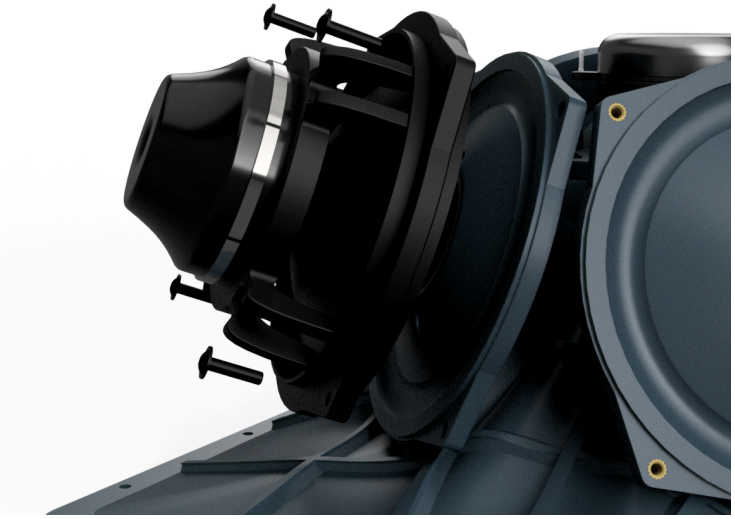
5. Attach the compression driver to the compression driver mounting flange using M5x12 countersunk fasteners. The compression driver mounting flange is self-aligning - tighten each fastener down gradually and evenly so that the exit aperture is centered on the mounting flange.



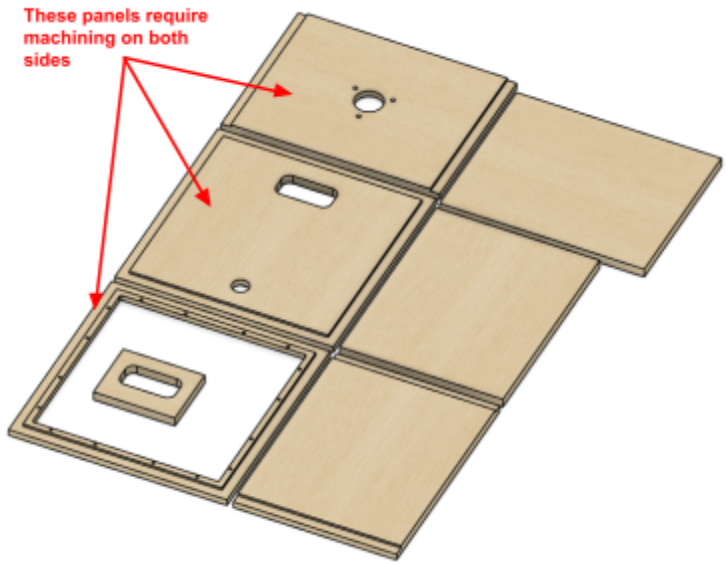
6. Fasten the compression driver assembly to the main waveguide body using 4x M4x12 fasteners.



7. Attach the four midrange drivers to the midrange mounting flanges using M5x20mm fasteners.

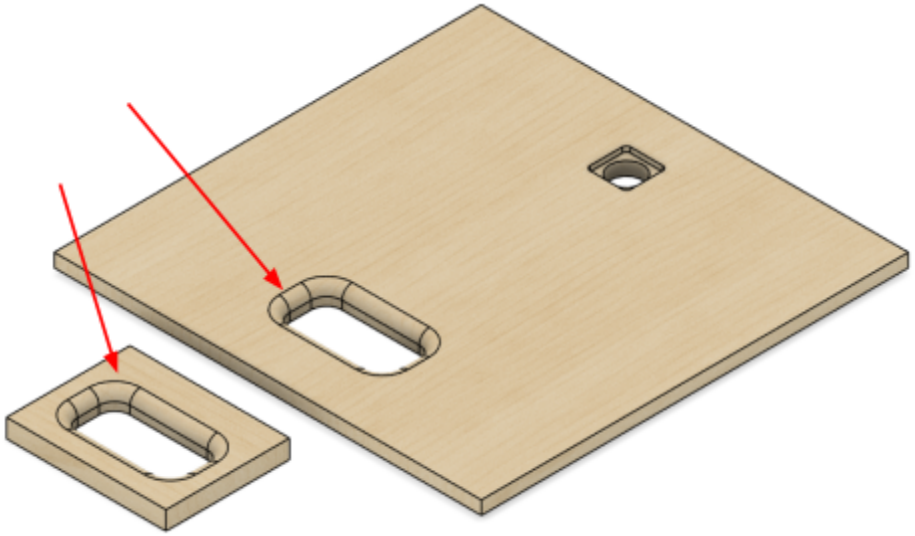


8. Cut all required enclosure panels from 18mm plywood. There are no 3d toolpaths or bevel cuts required. Enclosure files are provided in the download package in 3 formats:
    a. A .dxf containing panel outlines. All pockets are machined at a 5mm depth. The pocket on the front baffle where the waveguide occupies is machined at a 6.5mm depth.
    b. A 3d .STEP file containing each panel
    c. A parametric Fusion .f3d file containing each panel with timeline edits preserved.

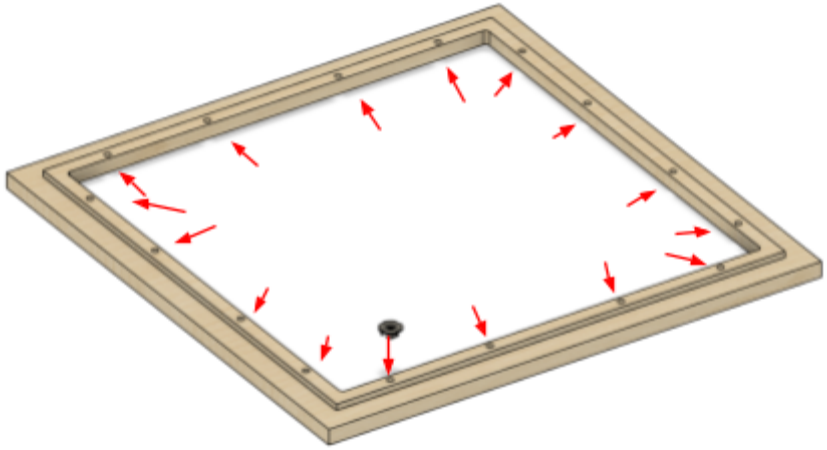


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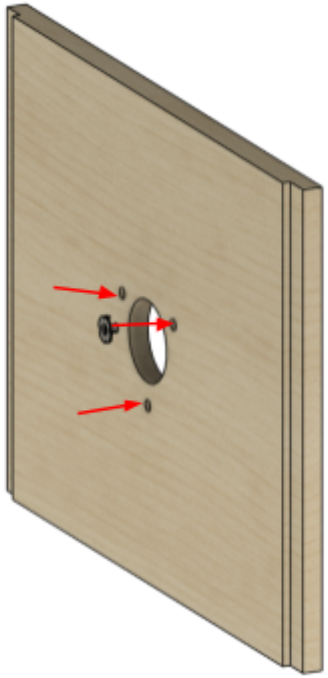
9. Using a 10mm (⅜”) roundover bit on a router, apply a fillet to the inside and outside edges of the port pieces.



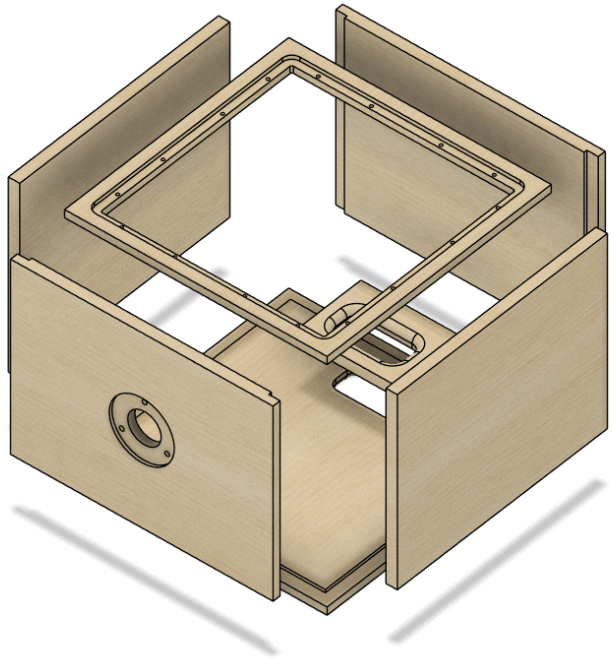
10. Insert 16x M4 tee nuts or furniture inserts into the baffle panel.



11. Insert 3x M6 tee nuts or furniture inserts into the bottom panel for the pole mount.



12. Assembly the enclosure using the appropriate adhesive, wood clamps, and brad nails or screws. Ensure that the port pieces are oriented with the rounded edges facing inwards/outwards to the enclosure. Line the interior of the enclosure walls with 2.5cm (1”) thick polyfill.



13. Wire the drivers as needed for your application and insert the waveguide module into the enclosure. Use M4x20mm button cap fasteners to attach the waveguide module to the enclosure with a ring of gasket tape. Install the speaker pole mount and NL4 receptacle. Wiring the woofers to pins 1+/- and the compression driver to pins 2+/- is recommended.



Changelog

Version	Release Date	Changes
1.0	2025-05-08	Initial 1.0 release
1.01	2025-05-06	Add enclosure simulation guidelines
1.02	2025-05-26	Add Klippel NFS results
1.03	2025-06-01	Refine enclosure simulation guidelines Add WinISD template
2.0	2026-08-31	2.0 redesign: - Modularize assembly/simplify construction - Improve waveguide directivity - Improve entry port geometry for less chuffing - Add high tuned enclosure plans